

Two-Step Gated GRPO for Long-Horizon Trajectories

Method note

1 Problem and data

Given old policy π_{old} , let τ_i have terminal reward R_i . A state s is a full decision prefix; u is its next action. Seek near-zero first advantages with clear successor-action separation.

2 Action-mean advantage at a state

$$\mathcal{I}(s) = \{i : \tau_i \ni s\}, \quad K_s = |\mathcal{I}(s)|, \quad \mathcal{U}(s) = \{a_i(s) : i \in \mathcal{I}(s)\}$$

$\mathcal{I}(s, u)$ groups trajectories taking u at s . Average within each action, then weight distinct actions equally; $d_s = |\mathcal{U}(s)|$.

$$q_s(u) = \frac{1}{n_s(u)} \sum_{i \in \mathcal{I}(s, u)} R_i, \quad n_s(u) = |\mathcal{I}(s, u)|$$

$$\bar{q}_s = \frac{1}{d_s} \sum_{u \in \mathcal{U}(s)} q_s(u), \quad \sigma_{q,s}^2 = \frac{1}{d_s} \sum_{u \in \mathcal{U}(s)} (q_s(u) - \bar{q}_s)^2$$

$$A_i^{\text{state}}(s) = \frac{q_s(a_i(s)) - \bar{q}_s}{\sigma_{q,s}} \quad (\sigma_{q,s} > 0)$$

Shared (s, u) gives shared A^{state} . Set zero-variance fallback advantage to 0; apply no inverse-count factor $1/n_s(u)$.

3 Two selection rounds and the Δ gate

$$\sigma_t = \sigma_{q, s_t}, \quad A_{1,i}(s_t) = A_i^{\text{state}}(s_t)$$

$$M_t = \lceil 0.2K_t \rceil, \quad C_t(s_t) = \{a_{i,t} : i \in \mathcal{L}_t(s_t)\}$$

Select M_t trajectories by smallest $|A_{1,i}|$ into L_t . For a tied action group crossing the cutoff, sample the remaining quota uniformly without replacement. Only selected trajectories can receive λ .

$$s_{t+1} = (s_t, a), \quad \mathcal{I}_{t+1} = \{i : \tau_i \ni s_{t+1}\}, \quad d = |\mathcal{U}(s_{t+1})| \geq 5$$

For candidate a , use all old trajectories at its successor. Reject zero parent/child action-mean deviation or fewer than five distinct successor a' .

$$\sigma_{t+1} = \sigma_{q, s_{t+1}}, \quad G_\Delta(s_t, a) = \mathbf{1}\{\sigma_{t+1} \geq \omega\sigma_t\}$$

$$A_2(s_{t+1}, u) = \frac{q_{s_{t+1}}(u) - \bar{q}_{s_{t+1}}}{\sigma_{t+1}}$$

$$m(d) = \lceil d/2 \rceil, \quad |S(s_t, a)| = m(d)$$

Require $\sigma_{t+1} \geq \omega\sigma_t$. Select $m(d)$ distinct a' with largest $|A_2|$ into S ; sample cutoff ties uniformly. Keep the advantage sign.

4 Gates and two-step advantages

$$G_t^{\text{act}}(s_t, a) = \mathbf{1}\{\sigma_t > 0\} \mathbf{1}\{a \in C_t(s_t)\} \mathbf{1}\{S(s_t, a) \neq \emptyset\}$$

$$B_{i,t} = \mathbf{1}\{i \in \mathcal{L}_t(s_t)\} G_t^{\text{act}}(s_t, a_{i,t})$$

A failed successor condition makes S empty, so $B=1$ implies a passing a' . Same- a trajectories outside L_t receive no λ .

$$\Delta A_{i,t}^{(1)} = \lambda B_{i,t}, \quad \lambda > 0$$

$$\widetilde{A}_{i,t+1} = [1 + (\beta - 1)I_{i,t+1}^S]A_2(s_{t+1}, a_{i,t+1}), \quad \beta > 1$$

For passing, non-overlapping successors, I^S marks membership in S . Here $A^{\text{state}}=A_2$: multiply unselected a' by 1 and selected a' by $\beta > 1$.

$$\widehat{A}_{i,j}^{\text{overlap}} = A_i^{\text{state}}(s_{i,j}) + \lambda + \beta I_{i,j}^S A_2(s_{i,j}, a_{i,j})$$

Overlap adds the next chain's $A^{\text{state}} + \lambda$ to the prior chain's $\beta I^S A_2$; if $I^S=1$, the action-level sum is $(1+\beta)A_2$.

5 Per-position advantage and GRPO update

$$P_{i,j} = G_{j-1}^{\text{act}}(s_{i,j-1}, a_{i,j-1}), \quad I_{i,j}^S = \mathbf{1}\{a_{i,j} \in S(s_{i,j-1}, a_{i,j-1})\}$$

The five cases define $\widehat{A}_{i,j}$; a nonexistent prior gate is zero at boundaries.

Condition	$\widehat{A}_{i,j}$
P=0, B=0	$A_i^{\text{state}}(s_{i,j})$
P=0, B=1	$A_i^{\text{state}}(s_{i,j}) + \lambda$
P=1, B=0, $I^S=1$	$\beta A_2(s_{i,j}, a_{i,j})$
P=1, B=0, $I^S=0$	$A_i^{\text{state}}(s_{i,j})$
P=1, B=1	$\widehat{A}_{i,j}^{\text{overlap}}$

$$r_{i,j}(\theta) = \frac{\pi_{\theta}(a_{i,j} | s_{i,j})}{\pi_{\text{old}}(a_{i,j} | s_{i,j})}$$

$$\ell_{\text{clip}}(r, A) = \min\{rA, \text{clip}(r, 1 - \varepsilon, 1 + \varepsilon)A\}$$

$$J(\theta) = \mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{j=1}^{T_i} \left(\ell_{\text{clip}}(r_{i,j}(\theta), \widehat{A}_{i,j}) - \kappa D_{i,j}^{\text{KL}}(\theta) \right) \right]$$

Expect over trajectory batches from π_{old} . Maximize $J(\theta)$ or minimize $-J(\theta)$. Fix old data, gates, and advantages within an update; differentiate current-policy ratios and optional KL only. Keep repeated-action trajectory terms without count correction. KL use and κ remain unspecified.

$$D_{i,j}^{\text{KL}} = \frac{\pi_{\text{ref}}(a_{i,j} | s_{i,j})}{\pi_{\theta}(a_{i,j} | s_{i,j})} - \log \frac{\pi_{\text{ref}}(a_{i,j} | s_{i,j})}{\pi_{\theta}(a_{i,j} | s_{i,j})} - 1$$

Optional DeepSeekMath sampled-action KL: ε is clip width, κ the KL weight, β the second-step amplifier.

6 Calculation order

Collect old trajectories $\rightarrow$ compute state/action means, deviations, and $A^{\text{state}} \rightarrow$ select the lowest 20% by trajectory $\rightarrow$ check successor $d \geq 5$, nonzero deviations, and the Δ gate $\rightarrow$ select $m(d)$ actions by $|A_2| \rightarrow$ assign $\widehat{A}$ and update θ .

This is a theoretical method specification. Numerical values of ω , λ , β and the mapping from actions to tokens or full steps remain unspecified. No experimental result is reported.

Original GRPO objective: Shao et al., DeepSeekMath, arXiv:2402.03300.